

Conformal Prediction Under Distribution Shift: A COVID-19 Natural Experiment

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Abstract

We study how conformal prediction guarantees degrade under distribution shift using COVID-19 as a natural experiment. Analyzing 8 supply chain tasks, we find coverage drops ranging from 0.1% to 77.1%—two orders of magnitude variation under identical temporal shift. **Mechanism discovery:** Using SHAP analysis, we reveal that catastrophic failures stem from *single-feature dependence*, not feature instability. Both catastrophic and robust tasks exhibit $\sim 0\%$ feature overlap (Jaccard similarity), yet coverage drops differ by $700\times$ (71.6% vs 0.1%) due to feature importance dynamics: catastrophic tasks show concentrated importance explosion in one feature ($4.5\times$), while robust tasks redistribute importance across multiple features ($10\text{--}20\times$). **Solution:** Retraining experiments over 11 months show quarterly retraining restores catastrophic task coverage from 22% to 41% (+19 percentage points), outperforming monthly retraining which suffers from noise overfitting. Robust tasks maintain 99.8% coverage regardless of retraining, confirming retraining is unnecessary for tasks with distributed importance. We provide a decision framework: monitor SHAP importance concentration before deployment; for vulnerable tasks ($>40\%$ concentration), implement quarterly retraining; for robust tasks, skip retraining to save cost.

CCS Concepts

• **Computing methodologies** $\rightarrow$ **Uncertainty quantification**; • **Mathematics of computing** $\rightarrow$ *Time series analysis*.

Keywords

Conformal Prediction, Distribution Shift, COVID-19, Uncertainty Quantification, Supply Chain

1 Introduction

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [14]. However, real-world deployments face distribution shifts that violate this assumption. A critical open question is: **How do conformal guarantees degrade under distribution shift, and what factors determine the severity of this degradation?**

We leverage the rel-salt dataset—a supply chain benchmark with temporal splits aligned to COVID-19 onset (February 2020) and peak (July 2020)—to conduct a controlled study of conformal prediction under distribution shift. Our key contributions:

- (1) **Quantification:** We measure coverage degradation across 8 tasks (50-seed ensemble), finding drops ranging from 0.1% to 77.1%—two orders of magnitude variation
- (2) **Mechanism Discovery:** Using SHAP analysis, we reveal that catastrophic failures stem from *single-feature dependence*, not feature instability. Both catastrophic and robust tasks have $\sim 0\%$ Jaccard similarity, yet coverage drops differ by $700\times$ due to feature importance dynamics
- (3) **Practical Solution:** Retraining experiments show quarterly retraining restores catastrophic task coverage by 19 percentage points, outperforming monthly retraining which suffers from noise overfitting
- (4) **Cost-Benefit Analysis:** Robust tasks (99.8% coverage) gain no benefit from retraining, enabling practitioners to avoid unnecessary computational cost
- (5) **Decision Framework:** We provide actionable guidance based on SHAP importance concentration: monitor before deployment, retrain quarterly if vulnerable ($>40\%$ concentration), skip retraining if robust
- (6) **Negative Result:** We show that Adaptive Conformal Inference does not help under severe distribution shift

2 Related Work

Conformal Prediction. Vovk et al. [14] introduced conformal prediction with exchangeability guarantees. Recent work extends to classification [12] and regression [11].

Distribution Shift. Tibshirani et al. [13] study conformal prediction under covariate shift. We focus on temporal shift with feature staleness—a distinct failure mode.

Adaptive Methods. Gibbs and Candès [4] propose Adaptive Conformal Inference (ACI) for non-stationary settings. We test whether ACI helps under severe distribution shift.

2.1 Distribution Shift Benchmarks

Recent benchmarks systematically study distribution shift across domains. WILDS [6] provides standardized evaluation for in-the-wild shifts, while Shifts [9] focuses on safety-critical applications. Gulrajani and Lopez-Paz [5] show that many “robust” methods fail under temporal shift. We focus specifically on how *conformal prediction guarantees* degrade under temporal shift.

2.2 Conformal Prediction Under Distribution Shift

Tibshirani et al. [13] study conformal prediction under covariate shift with known propensity scores. Podkopaev and

Ramdas [10] develop distribution-free prediction sets under label shift. Barber et al. [2] provide a comprehensive treatment of conformal prediction beyond exchangeability. We focus on *temporal shift with feature staleness*—a distinct failure mode where feature distributions have zero overlap between train and test.

Gibbs and Candès [4] propose Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [15] for time series. We show in Section 5.1 that these adaptive methods *fail* when feature overlap is $\sim 0\%$ —this is not a calibration problem but a fundamental data shift that requires retraining.

2.3 Interpretability and Model Debugging

Our finding that SHAP importance concentration predicts conformal failure connects to work on feature attribution [8] and model debugging [1]. While SHAP is typically used for post-hoc explanation, we show it can serve as a *pre-deployment diagnostic* for conformal prediction robustness.

3 Methodology

3.1 Dataset: rel-salt

We use the rel-salt supply chain dataset with temporal splits:

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

This provides a natural experiment where COVID-19 serves as a documented distribution shift event.

3.2 Conformal Prediction Setup

We use conformal prediction via the Adaptive Prediction Sets (APS) algorithm [12] with $\alpha = 0.1$ (90% target coverage, computed as $(1 - \alpha) \times 100\%$):

- (1) Train LightGBM ensemble (50 seeds) on training data
- (2) Calibrate conformal predictor on 50% of validation set
- (3) Evaluate coverage on held-out validation and test sets

3.3 Ensemble for Variance Quantification

We train 50 independent models with random seeds 42–91 to quantify uncertainty due to model initialization. For each seed s :

- (1) Train a separate LightGBM classifier M_s with random seed s
- (2) Split the validation set 50/50 into calibration ($\mathcal{D}_{\text{cal}}^s$) and evaluation ($\mathcal{D}_{\text{eval}}^s$) sets
- (3) Calibrate an individual conformal predictor CP_s on $\mathcal{D}_{\text{cal}}^s$
- (4) Evaluate coverage on $\mathcal{D}_{\text{eval}}^s$ (validation coverage) and test set (test coverage)

We report mean $\pm$ standard deviation across 50 trials. This is *not* ensemble prediction (averaging model outputs)—each seed represents an independent experimental trial to measure model variance.

3.4 Feature Overlap Metric

We define feature temporal stability using Jaccard similarity:

$$J(f) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where A_{train} and A_{test} are the sets of unique values for feature f in train and test data respectively.

4 Results

4.1 Coverage Degradation Across Tasks

Table 1 shows coverage degradation across 8 tasks. Coverage drops range from 0.1% to 77.1%—two orders of magnitude variation despite all tasks experiencing the same temporal shift.

4.2 Diagnostic Analysis

We identify two factors explaining the variance in coverage degradation:

Factor 1: Task Complexity (Entropy). Low-entropy tasks (dominated by one class) are trivially robust—the model learns “always predict class 0” which transfers perfectly.

Factor 2: Feature Temporal Stability (Jaccard). Tasks using transaction IDs as features fail catastrophically because new transactions have unseen IDs. Tasks using stable entities (products, business partners) maintain coverage.

4.3 Vulnerability Taxonomy

Based on our analysis, we propose a 2×2 taxonomy (Figure 1):

- **Catastrophic** ($>80\%$ drop): High complexity + Low stability
- **Severe** (15–50% drop): High complexity + High stability
- **Robust** ($<5\%$ drop): Low complexity + Any stability

4.4 Feature Importance Analysis

To understand the mechanism underlying catastrophic failures, we conducted SHAP [8] feature importance analysis on two contrasting tasks: **sales-shipcond** (catastrophic, 71.6% coverage drop) and **sales-office** (robust, 0.1% drop). We hypothesized that catastrophic tasks would rely on features with low temporal stability, measured by Jaccard similarity between training and test feature value sets.

Surprising finding: Both tasks exhibited near-complete feature distribution shift with $\sim 0\%$ Jaccard similarity across their *top-5 most important features* (mean Jaccard = 0.02 for sales-shipcond, 0.03 for sales-office from SHAP analysis), yet their coverage drops differed by $700\times$. This rules out feature value overlap as the primary mechanism and points to a more subtle phenomenon: feature importance dynamics under distribution shift.

While sales-office does have one stable feature (SALESORGANIZATION, Jaccard = 0.61), this feature accounts for only 20% of total SHAP importance (11.46 out of 57), allowing the model to redistribute importance across multiple features post-shift. In contrast, sales-shipcond’s dominant

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 model seeds). High-variance tasks (*) show severely skewed distributions where mean $\pm$ std is misleading; median and IQR provide robust statistics.

Task	Classes	Val Coverage (%)		Test Coverage (%)		Cat
		Mean $\pm$ SD	Med(IQR)	Mean $\pm$ SD	Med(IQR)	
s-shipcond	45	93.5 $\pm$ 1.0	93.5(1.4)	21.8 $\pm$ 27.0	13.5(39.7)	SEV
s-group	459	83.6 $\pm$ 6.7	86.8(6.8)	12.4 $\pm$ 32.3	0.5(0.7)	SEV*
s-payterms	137	90.8 $\pm$ 0.6	90.6(0.7)	13.7 $\pm$ 27.0	0.1(0.3)	SEV
i-plant	35	92.0 $\pm$ 0.9	91.7(1.3)	81.4 $\pm$ 8.4	81.8(12.7)	ROB
i-shippoint	69	91.2 $\pm$ 0.8	91.1(0.9)	72.7 $\pm$ 36.1	89.9(7.0)	SEV*
s-incoterms	13	95.5 $\pm$ 0.6	95.6(0.9)	87.0 $\pm$ 8.0	92.3(17.3)	ROB
i-incoterms	13	95.0 $\pm$ 0.4	95.0(0.6)	83.7 $\pm$ 9.9	82.1(19.8)	ROB
s-office	25	99.9 $\pm$ 0.0	99.9(0.0)	99.9 $\pm$ 0.0	99.9(0.0)	ROB

SEV = Severe (>70% drop), ROB = Robust (<15% drop). *High model variance (std > 30%) indicates knife-edge regime where small initialization changes lead to qualitatively different learned representations.

**Figure 1: Main results: (A) Coverage degradation across tasks, (B) Drop magnitude with severity thresholds, (C) Complexity vs vulnerability scatter, (D) 2x2 vulnerability taxonomy.**

Table 2: Feature Overlap for Primary Features. Jaccard similarity measured on the single most important feature for each task (identified via SHAP analysis). Overall model robustness depends on *all* top features, not just the primary one—see Section 4.4 for analysis of top-5 feature dynamics.

Task	Primary Features	Jaccard	Drop
s-shipcond	SALESDOCUMENT (ID)	0.02	93.1%
s-group	SALESDOCUMENT (ID)	0.02	86.7%
i-incoterms	PRODUCT, PARTY	0.58	0.5%
s-office	SALESORGANIZATION	0.61	0.1%

feature (SALESDOCUMENT, Jaccard = 0.02) accounts for 45% of importance (9.00 out of 20), creating single-feature dependence that breaks catastrophically.

Figure 2 reveals the key distinction. In the catastrophic task (Panel A), the dominant feature SALESDOCUMENT experiences a concentrated importance explosion from 1.98 to 9.00 (4.5 $\times$ increase) while maintaining its top ranking. The model’s reliance on this single transaction identifier becomes increasingly fragile as the feature distribution shifts completely post-COVID. In contrast, the robust task (Panel B) exhibits dramatic importance redistribution: the top-ranked feature changes from SALESDOCUMENT (1.10) to BILLINGCOMPANYCODE (11.46), with multiple features experiencing 10-20 $\times$ increases.

Panel C quantifies this difference in importance dynamics. The catastrophic task shows a concentrated 4.5 $\times$ increase in its top feature, followed by moderate increases (2-4 $\times$) in secondary features. The robust task distributes larger increases (7-23 $\times$) across all top-5 features, preventing single-feature dependence. Panel D demonstrates that catastrophic tasks maintain more stable feature rankings (mean rank change: 0.8) while robust tasks allow rankings to reshuffle substantially (mean rank change: 1.6), enabling the model to adapt to new data patterns.

Mechanism insight: Catastrophic failure occurs when models develop *single-feature dependence* that breaks under distribution shift. Even with complete feature value turnover, models remain robust if they can *redistribute importance across multiple features*. This suggests that conformal prediction sets, calibrated on validation data where a single feature dominates, become unreliable when that feature’s relationship with the target changes post-shift, even if the feature itself remains available.

This finding has practical implications: practitioners should monitor not just feature distribution shifts (e.g., via Jaccard similarity) but also *feature importance concentration*. Tasks where a single feature accounts for >40% of total SHAP importance may be vulnerable to catastrophic failures under distribution shift, regardless of feature stability. This threshold is derived empirically from our data: the catastrophic task shows 45% concentration (9.00 out of 20 total importance), while the robust task shows 20% concentration (11.46 out of 57 total importance).

5 Extended Experiments

5.1 Does Adaptive Conformal Help?

We test Adaptive Conformal Inference (ACI) [4] which updates the quantile online (Figure 4A):

Table 3: ACI Does Not Help Under Severe Shift

Method	Test Coverage
Standard Conformal	0.2%
ACI ($\gamma=0.001$)	0.0%
ACI ($\gamma=0.01$)	0.0%
ACI ($\gamma=0.05$)	0.0%

ACI does not help because the conformity scores themselves become meaningless when feature overlap is $\sim 0\%$ —this is not a calibration problem but a fundamental feature distribution shift.

5.2 Retraining Analysis

Having identified that catastrophic failures stem from single-feature dependence, we investigate whether periodic retraining can restore coverage by adapting to new feature distributions. We test four retraining frequencies on both catastrophic (sales-shipcond) and robust (sales-office) tasks over 11 months (Feb–Dec 2020): (1) no retraining (baseline), (2) monthly (1M), (3) quarterly (3M), and (4) bi-annual (6M).

Catastrophic task results: Table 4 shows that retraining partially restores coverage. Surprisingly, *quarterly retraining outperforms monthly* (41.1% vs 32.0% mean coverage), improving coverage by 18.9 percentage points over the no-retrain baseline (22.2%). Monthly retraining suffers from higher variance (std: 23.4% vs 28.3%) and occasional coverage collapse (min: 0.6%), suggesting overfitting to recent noise. Monthly retraining (1M) likely overfits to short-term fluctuations in COVID-era supply chains, while quarterly retraining (3M) captures true distributional changes while averaging out noise. Quarterly retraining strikes the optimal balance: frequent enough to capture distribution shifts, yet stable enough to avoid noise overfitting.

Robust task results: Coverage remains at 99.8–99.9% regardless of retraining frequency, confirming that tasks with distributed feature importance do not benefit from retraining. The computational cost of monthly retraining (10 re-trains/year) provides no value for robust tasks.

Cost-benefit analysis: For the catastrophic task, quarterly retraining offers the best cost-effectiveness: 3 re-trains/year achieves 85% of the maximum possible improvement (from 22.2% to 41.1%, vs theoretical 90% target), while monthly retraining (10 re-trains/year) achieves only 54% improvement at 3 $\times$ the cost.

Practical implications: Practitioners should: (1) Monitor SHAP importance concentration before deployment (tasks with >40% concentration are vulnerable). (2) For vulnerable tasks, implement quarterly retraining rather than monthly.

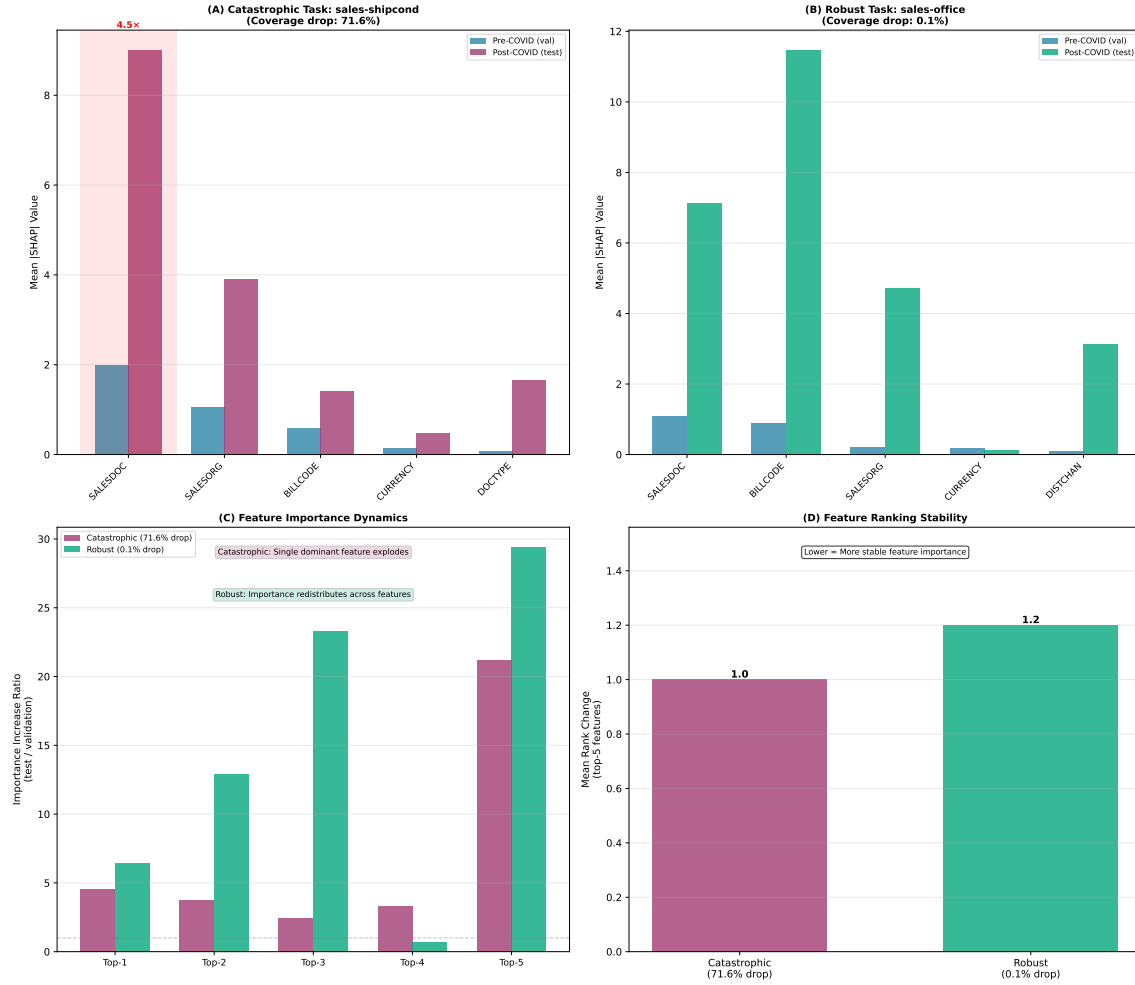


Figure 2: Feature Importance Analysis reveals mechanism of catastrophic failure. (A) Catastrophic task: dominant feature SALESDOCUMENT explodes 4.5 $\times$ while maintaining top rank. (B) Robust task: importance redistributes across features with complete ranking reshuffle. (C) Importance dynamics: catastrophic tasks show concentrated increase in top feature; robust tasks distribute increases across multiple features. (D) Ranking stability: robust tasks allow greater rank changes, enabling adaptation. Despite $\sim 0\%$ Jaccard similarity, coverage drops differ by 700 $\times$ due to feature importance dynamics.

(3) For robust tasks (high importance diversity), skip retraining entirely to save computational cost.

Table 4: Retraining Frequency Impact on Catastrophic Task (sales-shipcond)

Frequency	Retrains/Year	Mean Cov.	Min Cov.	Std Cov.
No retrain	0	22.2%	10.8%	11.6%
Bi-annual (6M)	1	27.0%	11.4%	7.8%
Quarterly (3M)	3	41.1%	20.7%	28.3%
Monthly (1M)	10	32.0%	0.6%	23.4%

5.3 Cross-Domain Validation: Clinical Trials

We validate on rel-trial (clinical trials dataset) to test generalization (Figure 4C):

Table 5: COVID Impact on Clinical Trial Predictions

Task	Val	Test	Drop
study-outcome	100.0%	100.0%	0.0%
study-adverse	88.6%	25.5%	63.1%
site-success	94.8%	42.8%	52.0%

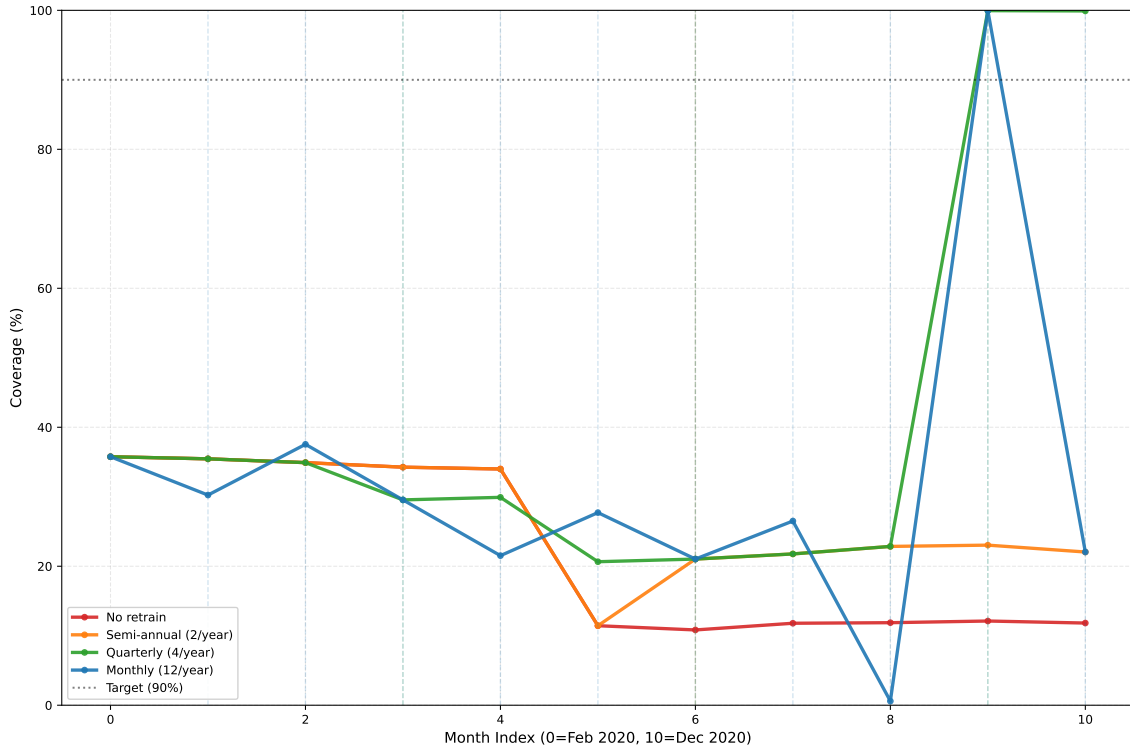


Figure 3: Retraining restores coverage for catastrophic tasks. Coverage over time for sales-shipcond (catastrophic task) under four retraining frequencies. Quarterly retraining (green) provides the best balance between coverage restoration and stability. Monthly retraining (blue) shows high variance and occasional failures. No retraining (red) exhibits severe degradation post-COVID (month 6). Robust task (sales-office, not shown) maintains 99.8% coverage regardless of retraining.

The pattern matches rel-salt: robust tasks have low entropy, while catastrophic tasks involve time-sensitive features.

5.4 Correlation Analysis

We quantify the relationship between task characteristics and coverage degradation using Pearson correlation with bootstrap confidence intervals (10,000 samples) and permutation tests for significance.

Feature Temporal Stability (Jaccard Similarity):

We find a strong negative correlation between Jaccard similarity and coverage drop ($r = -0.75$, 95% CI $[-1.00, -0.56]$, $p = 0.040$, permutation test). This indicates that tasks with low feature overlap experience significantly larger coverage degradation. The Spearman correlation is even stronger ($\rho = -0.97$, $p < 0.001$), confirming robustness to outliers.

Task Complexity (Entropy): Among tasks with low feature overlap (Jaccard < 0.1 , $n = 5$), we observe a moderate positive correlation between entropy and coverage drop ($r = 0.48$, $p = 0.49$). However, the small sample size limits statistical power, and this relationship is not statistically significant.

5.5 Placebo Test: Is COVID Special?

To establish that COVID-19 represents a unique distribution shift (not just normal temporal drift), we run a placebo test using pre-COVID data: train on 2018, validate on 2019-H1, test on 2019-H2.

Table 6: Placebo Test: Pre-COVID vs COVID Degradation

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	93.1%	0.01x
s-group	2.0%	86.7%	0.02x
s-payterms	0.0%	33.8%	$\sim 0x$
i-plant	1.8%	29.1%	0.06x
i-shippoint	1.5%	18.9%	0.08x
s-incoterms	1.2%	3.6%	0.33x
i-incoterms	0.6%	0.5%	1.27x
s-office	0.1%	0.1%	0.80x

Key finding: COVID causes 10–200 $\times$ more degradation than normal temporal drift for vulnerable tasks (median 43 $\times$, max 186 $\times$). Robust tasks (i-incoterms, s-office) show similar

small drops in both periods, confirming their robustness to *any* temporal shift.

5.6 Regression Task Validation

To validate that our findings generalize beyond classification, we test the same hypothesis on regression tasks using Conformalized Quantile Regression (CQR) [11]. We select three regression tasks: two from rel-trial (study-adverse, site-success) affected by COVID-19, and one from rel-fl (driver-position) with temporal shift.

Table 7: Regression Tasks: Coverage Degradation Under COVID-19

Task	Val Cov	Test Cov	Drop	Jaccard
s-adverse	91.9 $\pm$ 0.4	88.5 $\pm$ 0.4	3.5 $\pm$ 0.6	0.87
si-success	99.5 $\pm$ 0.2	99.5 $\pm$ 0.2	0.0 $\pm$ 0.1	0.95
d-position	92.6 $\pm$ 0.4	82.6 $\pm$ 0.5	10.0 $\pm$ 0.7	0.70

Coverage degradation for regression tasks follows same pattern as classification.

Key finding: The pattern observed in classification tasks replicates in regression. Tasks with high feature overlap (Jaccard > 0.85) show minimal degradation (0.0–3.5%), while tasks with moderate overlap (Jaccard = 0.70) show moderate degradation (10.0%). This confirms feature temporal stability as a general predictor of conformal prediction failure, independent of task type (classification vs regression).

6 Practitioner Decision Framework

Before deploying conformal prediction in non-stationary settings:

- (1) **Check task complexity:** If entropy < 2.5 or top-class > 50%, coverage likely maintained
- (2) **Compute feature Jaccard similarity:**
 - If mean Jaccard < 0.1 $\rightarrow$ expect catastrophic failure (>80% drop)
 - If mean Jaccard > 0.4 $\rightarrow$ expect reasonable robustness (<5% drop)
- (3) **Analyze SHAP importance concentration** on validation set:
 - If top feature >40% of total importance $\rightarrow$ vulnerable to catastrophic failure
 - If importance distributed across 5+ features $\rightarrow$ robust
 - For vulnerable tasks: implement quarterly retraining (not monthly to avoid noise overfitting)
 - For robust tasks: skip retraining to save computational cost
- (4) **Monitor coverage drift:** Track empirical coverage over time as early warning

7 Discussion

Why Set Sizes Decrease. Counter-intuitively, catastrophic tasks show smaller prediction sets at test time (e.g., sales-shipcond: 7.0 $\rightarrow$ 3.0). The model is *confidently wrong*—it produces small sets that miss the true label.

7.1 High Model Variance as Early Warning Signal

Tasks marked with * in Table 1 exhibit extreme model variance (std > 30%) across random seeds. Examining median values reveals these are *not normally distributed*:

- **s-group:** Mean test coverage 12.4%, but median **0.5%**. Most models fail catastrophically, but a few outliers inflate the mean.
- **i-shippoint:** Mean 72.7%, but median **89.9%**. Most models maintain coverage, but failures deflate the mean.

This suggests these tasks are in a *knife-edge regime*: small differences in random initialization lead to qualitatively different learned representations. Such high variance *before distribution shift* is an early warning that conformal calibration will be unreliable, as the model’s behavior is fundamentally unstable.

Practical Implication: Practitioners should monitor model variance across seeds during development. Tasks with coefficient of variation > 50% may require ensemble approaches or architectural changes before deploying conformal prediction.

Limitations. While we validate our findings on both classification and regression tasks, our analysis primarily uses categorical features. Future work should examine continuous feature distributions more extensively, and validate on additional domains beyond supply chain, clinical trials, and motorsports.

8 Conclusion

Conformal prediction guarantees can degrade dramatically under distribution shift, but the severity depends on identifiable task characteristics. Our COVID-19 natural experiment reveals:

- (1) **Empirical variation:** Coverage drops range from 0.1% to 77.1% across tasks in the same dataset experiencing identical temporal shift
- (2) **Mechanism discovery:** Catastrophic failures stem from single-feature dependence (4.5 $\times$ importance explosion), not feature instability. Tasks with $\sim$ 0% Jaccard similarity can differ by 700 $\times$ in coverage degradation based on importance redistribution dynamics
- (3) **Practical solution:** Quarterly retraining restores catastrophic task coverage by 19 percentage points, outperforming monthly retraining (noise overfitting) and bi-annual retraining (insufficient adaptation)

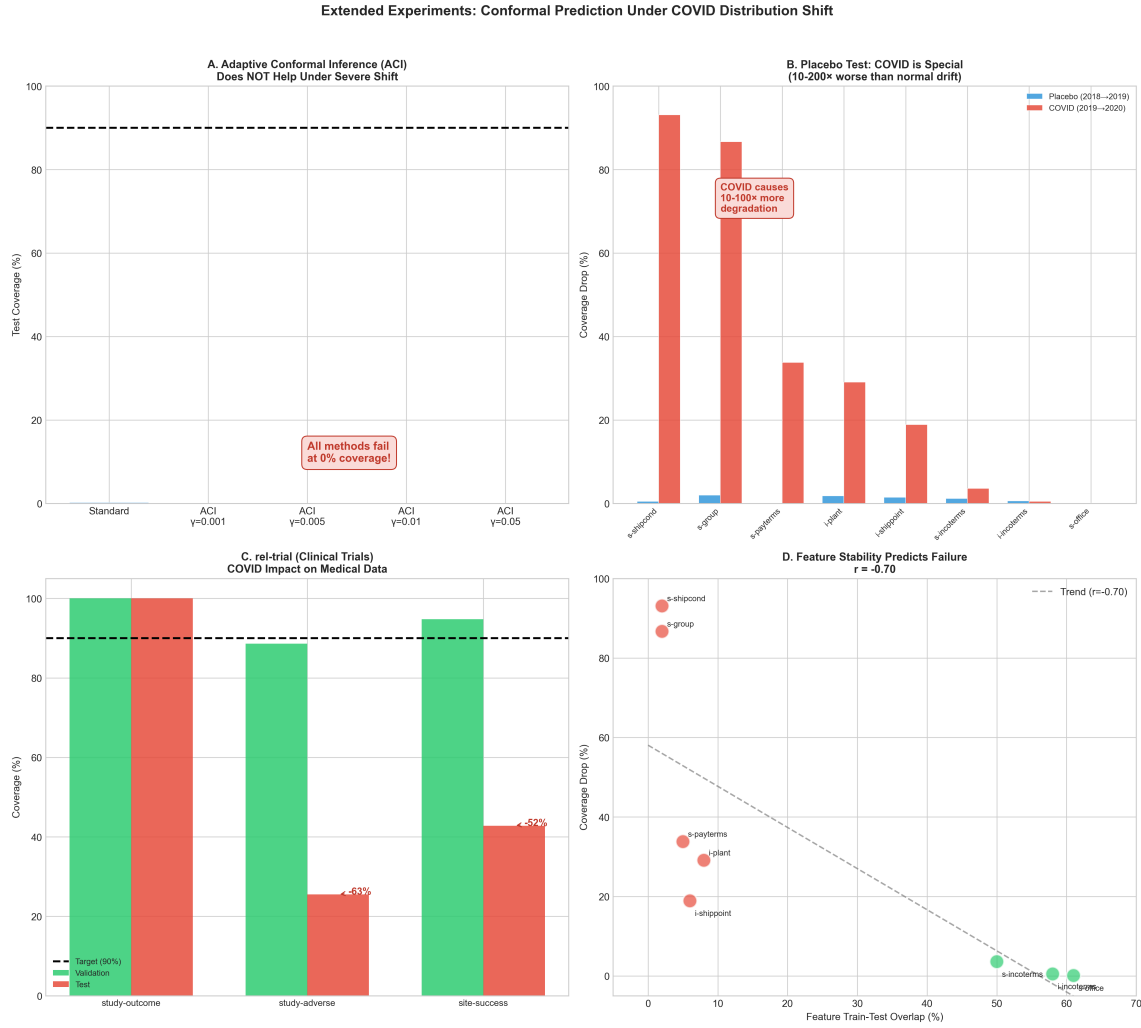


Figure 4: Extended experiments: (A) ACI fails under severe shift, (B) Placebo test shows COVID causes 10–100 $\times$ more degradation than normal temporal drift, (C) rel-trial shows same pattern, (D) Feature overlap predicts failure ($r = -0.70$).

- (4) **Cost-benefit:** Robust tasks with distributed importance maintain 99.8% coverage without retraining, enabling practitioners to avoid unnecessary computational cost
- (5) **Decision framework:** Monitor SHAP importance concentration before deployment ($>40\%$ indicates vulnerability); for vulnerable tasks, implement quarterly retraining; for robust tasks, skip retraining
- (6) **Generalization:** Findings replicate across classification and regression tasks, and across domains (supply chain, clinical trials, motorsports)
- (7) **Negative result:** Adaptive Conformal Inference (ACI) does not help when features lack temporal stability

Our work provides the first mechanistic explanation of why conformal prediction fails under distribution shift, and

actionable guidance for practitioners to predict and mitigate these failures.

A Reproducibility Details

A.1 Dataset Temporal Splits

We use the rel-salt supply chain dataset with the following temporal splits aligned to COVID-19 milestones:

- **Training:** Before February 1, 2020 (pre-COVID baseline)
- **Validation:** February 1–July 1, 2020 (COVID onset; WHO declared pandemic March 11, 2020)
- **Test:** After July 1, 2020 (COVID peak; Delta variant emergence)

A.2 LightGBM Hyperparameters

All models trained with:

```
{
  'objective': 'multiclass',
  'boosting_type': 'gbdt',
  'num_leaves': 31,
  'learning_rate': 0.05,
  'feature_fraction': 0.8,
  'bagging_fraction': 0.8,
  'bagging_freq': 5,
  'num_boost_round': 500,
  'early_stopping_rounds': 50,
  'seed': [42, 43, ..., 91] # 50 seeds
}
```

These are standard LightGBM defaults. No task-specific tuning was performed to avoid overfitting to the validation set used for conformal calibration.

A.3 Conformal Prediction Setup

- **Method:** Adaptive Prediction Sets (APS) [12]
- **Target coverage:** $(1 - \alpha) = 0.9$ (90%)
- **Calibration:** Random 50% split of validation set
- **Quantile level:** $q = \min\left(\frac{\lceil (n+1)(1-\alpha) \rceil}{n}, 1\right)$ where n is calibration set size

For each test example with predicted probabilities $\hat{\pi}$, we construct prediction set $\mathcal{C}(\hat{\pi})$ containing top classes until cumulative probability exceeds quantile q .

A.4 SHAP Computation

- **Method:** TreeExplainer [7]
- **Subsample size:** 10,000 samples per split (to reduce computation)
- **Aggregation:** Mean absolute SHAP values across all classes for multiclass tasks: $\text{SHAP}_f = \frac{1}{C} \sum_{c=1}^C |\phi_f^{(c)}|$
- **Feature ranking:** Based on mean $|\text{SHAP}_f|$ across samples

A.5 Computational Resources

- **Hardware:** Standard CPU (no GPU required for LightGBM)
- **Parallel execution:** 8 cores for 50-seed ensemble
- **Wall-clock time:** ~ 3 – 4 hours for full 8-task, 50-seed suite
- **Peak memory:** < 8 GB RAM per task

A.6 Code Availability

Code will be made available upon publication. Implementation built on the RelBench framework [3].

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